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NAME: Sabnish Seetharaman STD.: II year SEC.: D ROLL NO.: 206 SUB.: Software Construction

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Experiment 1 - Azure Devops Environment Setup

Aim :

To setup and acces the azure devops environment by creating an organization through the azure portal.

Installation :

1. Open your web browser and go the the azure Website.

https://azure.microsoft.com/en-us/get-started/azure-portal

sign in your using your microsoft credentials.

If you dont have a microsoft account, create one there:

https://signup.live.com/?lic=1

| Get started with Azure | | sign in |

2. Azure Home Page

Azure Services - Azure Devops Organizations.

3. Open Devops environment in the Azure platform by trying Azure Devops Organizations in the search bar.

4. Click on the My Azure Devops Organization link and create an organization and you should be taken to the Azure Devops Organization Home page.

Result :

Successfully accessed the Azure Devops environment and created a new organization through the Azure portal.

Experiment 2: Azure Devops Project Setup and User Story Management

Aim:

To setup on Azure Devops project for efficient & collaboration and agile work management.

Procedure:

1. Create an Azure account.

   Azure Devops

   sabnish1776@gmail.com.

   Name your Azure Devops Organization.

   dev.azure.com/ sabnish1776

2. Create the first project in your organization.

   (a) After the organization is setup, you'll need to create your first project. This is where you'll begin to manage code, pipelines, works items and more

   (b) On the organizations home page, click on the new project button.

   (c) Enter the project name, description and visibility options

   name: choose a name for the project.

   Description: Optionally, add a description to provide context about the project.

(4) Project Dashboard.

About this Project..

The Cultural Carnival management system is a cloud based application designed to efficiently, plan, organize, and manage large scale cultural events, built using Azure Devops and deployed on Microsoft Azure, it streamlines events coordination by integrating ~~mag~~ management and resource allocation on a single platform.

(5) To manage User Stories.

(a) From the left hand navigation menu, click on boards. This will take you to the main Boards page, where you can manage work items, backlogs and sprints.

(b) On work Items page you'll see the options to add a word item at the top. Alternatively, you can find a + button or Add New Work Item depending on view youre in - From the add dropdown, select User Stories. This will open a form to enter details for the new user story.

Boards → work items → + new work Item → Userstory

Result:
Successfully created an Azure Devops project with user story management & agile workflow setup.

visibility : choose whether your want the project to be private or public.

(d) Once you've filled about the details, click create, step your project.

→ Create New Project

project Name : Cultural Carnival Management System.

visibility : Public

Version Control : Git

Work ~~Item~~ process : Agile.

(3) Once logged in, ensure you are in the correct organization. If you're a part of multiple organizations you can switch between them from the top left corner (next to your user project). Click on the organization name and you should be taken to the Azure Devops organization Home page.

SS

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sabnish 1776 @gmail.com

projects : Cultural Carnival Management System

Exp. 3 Setting up epics, features and user stories for project planning.

Aim:

To learn about the to create epics, user story, features, backlogs for your assigned project.

Procedure:

(1.) File in Epics

(i) Enable students and organizers to securely access the platform and manage their profiles.

(ii) Provide a system to create events and allow students to register for them efficiently.

(2) Fill in features.

(i) Secure Login & Authentication

(ii) Student Account Registration.

(iii) Event Creation & management

(iv) Event Browsing & Filtering

(v) Event Registration System.

(vi) Role Based Access Control.

(3) Fill in User Story titles.

(i) User Registration.

(ii) Login Interface (UI + validation)

(iii) Registration confirmation.

(iv) Authentication System (JWT)

(v) Role Based Access control.

(vi) Role Based Dashboard.

(vii) Event Creation & Editing (admin UI)

(viii) Event CRUD System.

(ix) Event Listing UI with Filters.

(x) Event Filtering Backend Logic

(xi) Event Registration Button + UX

(xii) Registration Processing.

Result:

Thus, the creation of epics, features & user story & task has been created successfully.

Sprint 2

(7) As an admin, I want an easy-to-use interface to create and edit events so that I can manage event details efficiently.

(8) As a system, I want to store, update and delete event data accurately so that the event information remains constant.

(9) As a students, I want a visually organized event listing with filters so that I can quickly find events. ~~based o~~

Sprint 3

(10) As a System, I want to retrieve and filter events based on user selection so that only relevant events are displayed.

(11) As a System, I want to process and store event registrations while preventing duplicates so that participation data remains accurate.

(12) As a Student, I want a simple registration button and clear confirmation messages so that I can easily register.

Result:

The ~~Sprints~~ are created for the online carnival cultural management System.

Experiment 4 - Sprint Planning.

Aim :

To assign user story to specific for cultural carnival management project.

Sprint Planning :

Sprint 1

→ Assigned User Stories for Sprint 1

(1) As a user/student, I want to register using my email so that I can access the platform & participate in events.

(2) As a student, I want to receive a confirmation after registration so that I know my account has been successfully created.

(3) As a user, I want a simple login interface with input fields and error messages so that I can easily login in my account.

(4) As a system, I want to authenticate user credentials securely so that only authorized users can access the platform.

(5) As a student/admin, I want to see a dashboard tailored to my roles so that I can easily access relevant features.

(6) As a system, I want to restrict access based on user roles so that users can only perform actions permitted to them.

Experiment 5 : Poker Estimation.

Aim :

To create poker estimation for the user stories - for cultural carnival management System

Poker Estimation

Story points measure :

- efforts
- complexity
- Time

Uses Fibonacci series : 1, 2, 3, 5, 8, 13, ...

(1) User Registration

Story points : 5

Reason : Includes validation, saving user, password encryption, and basic error handling. Moderate complexity.

(2) Login Interface (UI + validation)

Story points : 3

Reason : Frontend + simple backend call. No heavy logic, just form handling & error message

(3) Registration confirmation

Story points : 2

Reason : Just success message + UI message. Low complexity.

(4) Authentication System (JWT/Session)

Story points: 8

Reason: Core security features + token generation, validation, filters, protected routes, High importance + complexity.

(5) Role Based Access control.

Story points: 5

Reason: Need to restrict endpoints + different UI views. Medium complexity with security logic.

(6) Role Based Dashboard.

Story points: 3

Reason:

Mostly frontend rendering based on Role. Backend impact is minimal.

(7) Event Creation & Editing.

Story points 5

Reason: CRUD operations + form + validation. moderate backend and frontend work.

(8) Event CRUD system. €

Story points: 5

Reason: Database operations, API design, consistency handing.

(9) Event listing UI with filters.

Story points: 3

Reason:

Frontend heavy with simple filtering logic.

(10) Event Filtering Backend Logic

Story points : 3
Reason : Query params + DB filtering. straight forward

(11) Event registration Button + UX

Story points : 2

Reason : simple UI trigger + API call.

(12) Registration processing for events

Story points : 5
Reason :
Needs validation (duplicate check), DB constraints
and proper User-Event Linking.

Result :

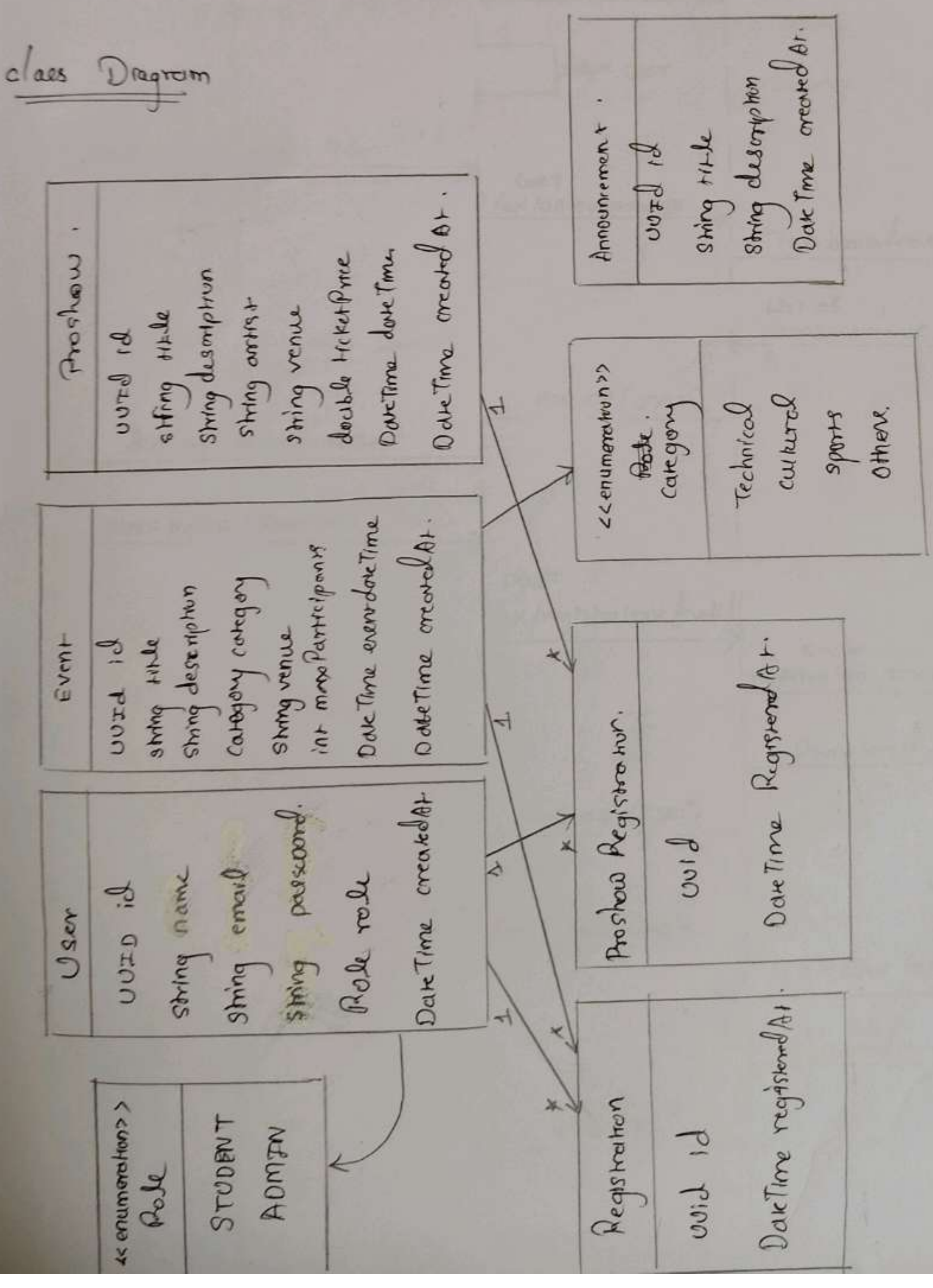
The poker Estimation for the user stories for cultural Carnival management System is done successfully.

Exp. 6 : Designing Class & Sequence Diagrams for project Architecture

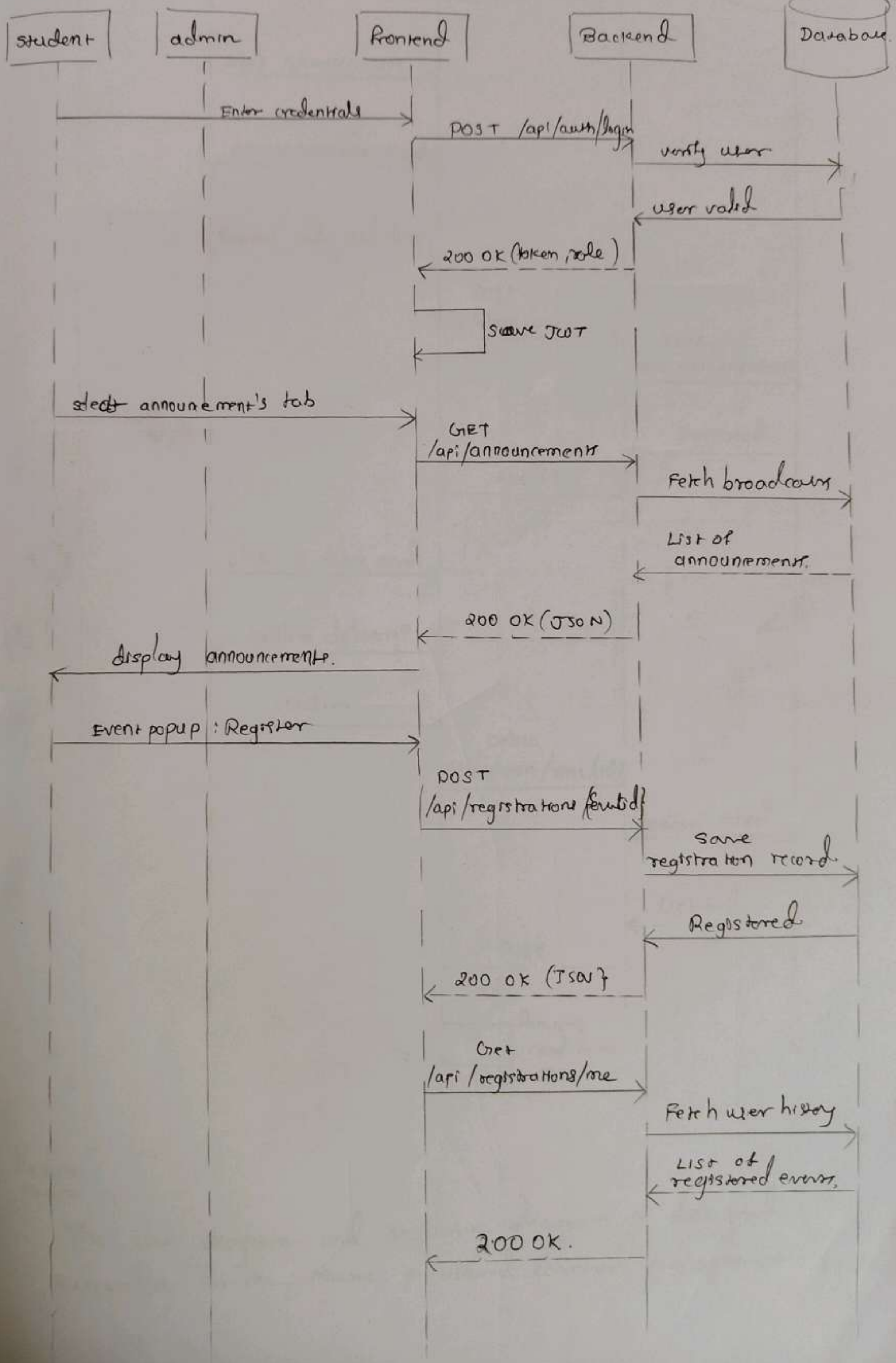
Aim:

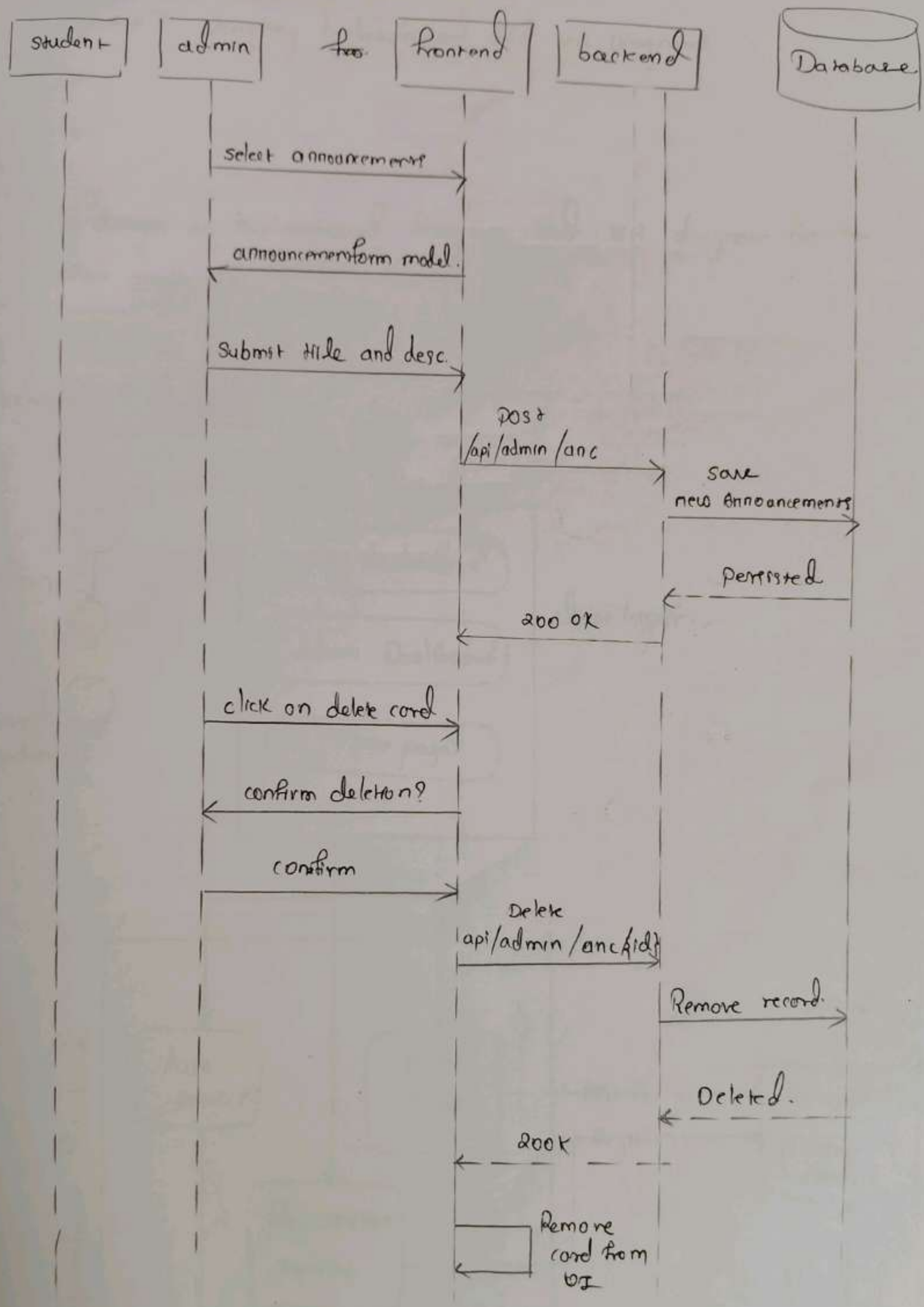
To design a class Diagram and sequence Diagram, for the project.

class Diagram



Sequence Diagram.





Result

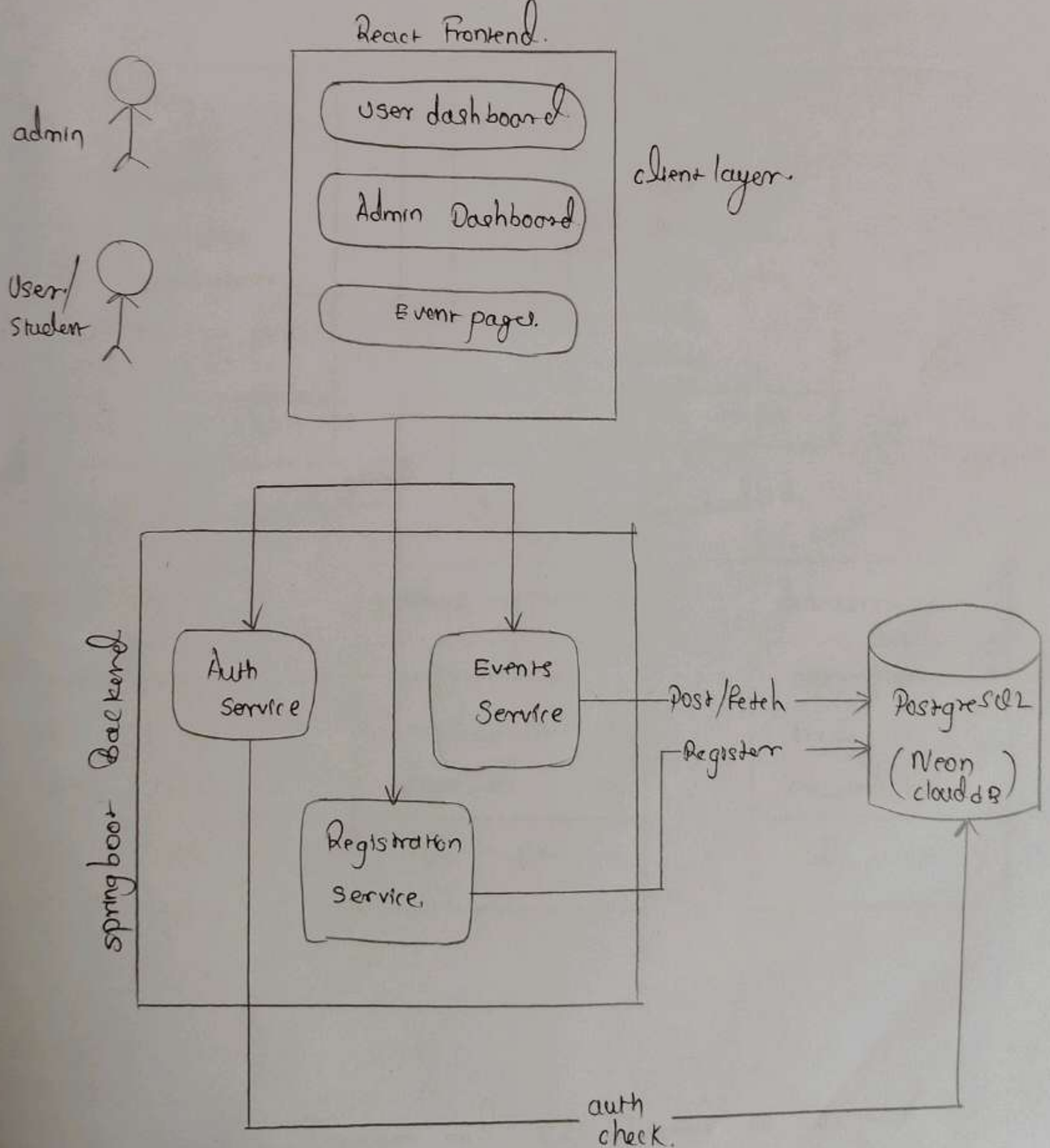
The class diagram and sequence diagram is designed successfully for the ~~Music~~ cultural carnival management system.

Exp no. 7 Designing Architectural and ER Diagrams

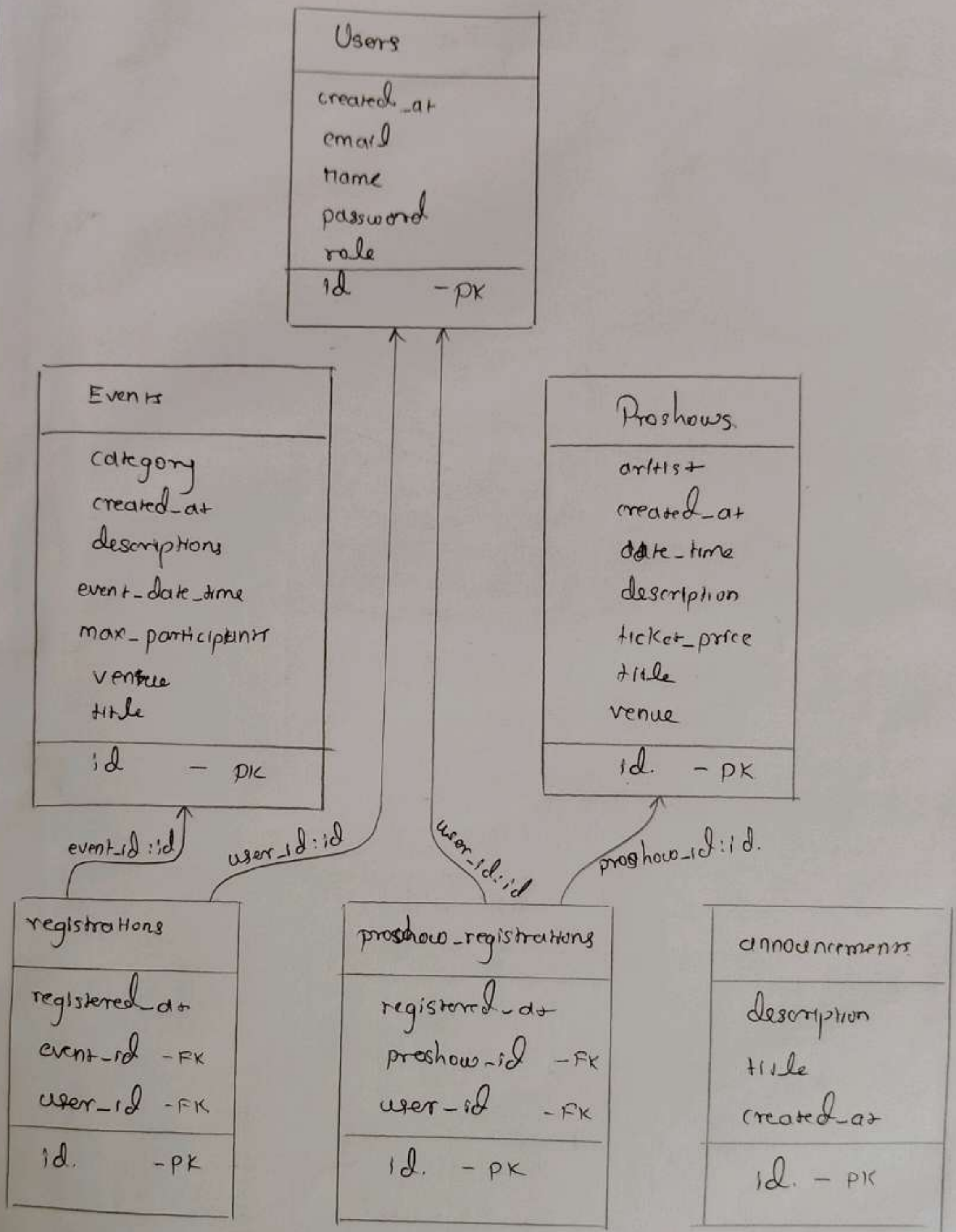
Aim :

To design an Architectural Diagram and ER diagram for the given project.

Architecture diagram.



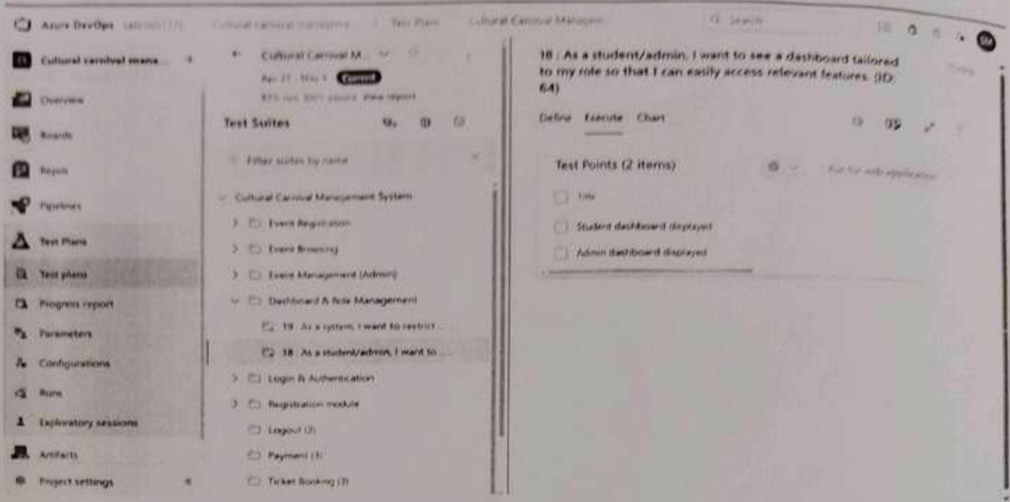
ER Diagram

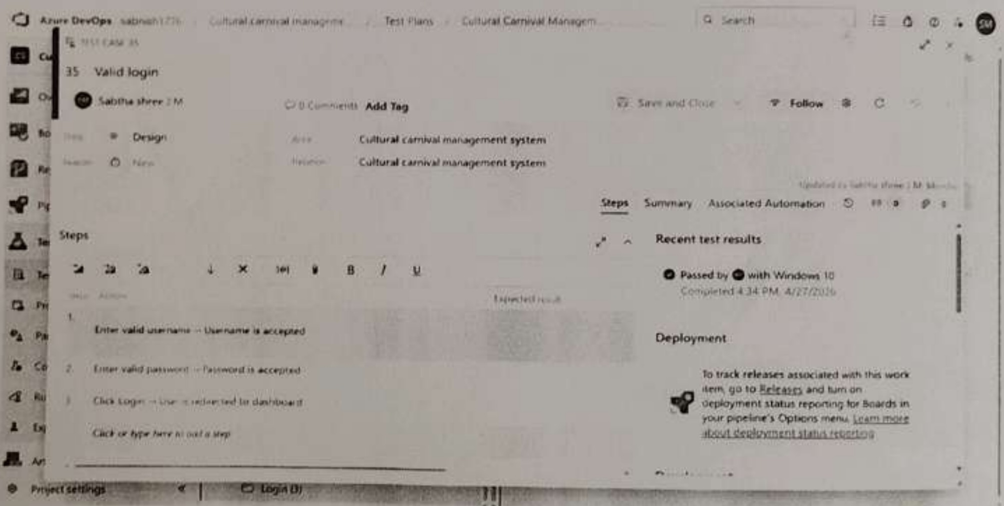


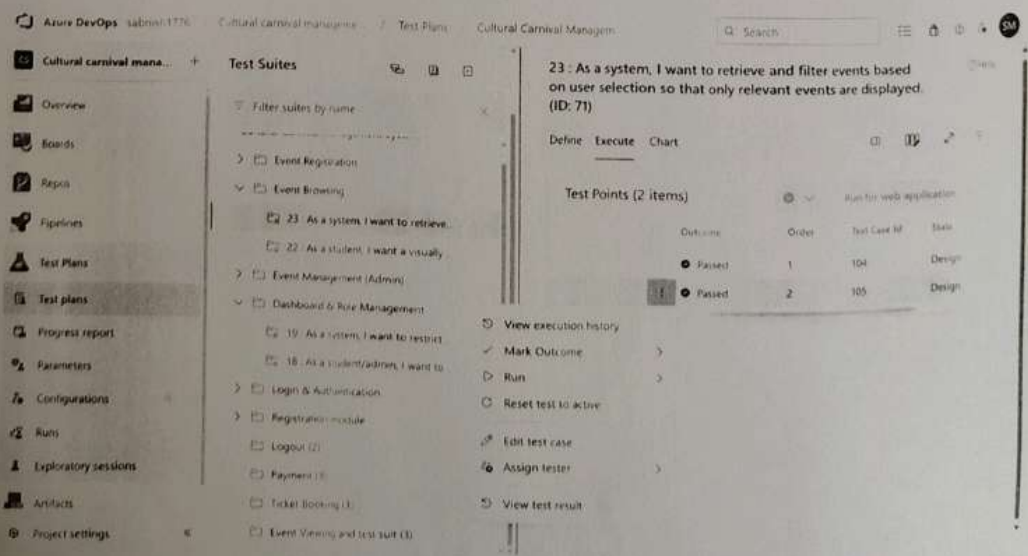
Result

The Architecture Diagram and ER Diagram is designed successfully for the cultural carnival management system.

TEST PLANS:


Azure DevOps
Test Suites
18 : As a student/admin, I want to see a dashboard tailored to my role so that I can easily access relevant features. (ID: 64)
Define
Execute
Chart
Test Points (2 items)
Student dashboard displayed
Admin dashboard displayed
Cultural Carnival Management System
Dashboard & Role Management
Login & Authentication
Registration module
Project settings


Azure DevOps
35 Valid login
Add Tag
Follow
Design
Cultural carnival management system
Steps
Summary
Associated Automation
Recent test results
Enter valid username → Username is accepted
Enter valid password → Password is accepted
Click Login → User is redirected to dashboard
Click or type here to add a step
Deployment
To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu.


Azure DevOps
Cultural Carnival Managem
Test Suites
23 : As a system, I want to retrieve and filter events based on user selection so that only relevant events are displayed. (ID: 71)
Define
Execute
Chart
Test Points (2 items)
Passed
Passed
Event Registration
Event Browsing
Event Management (Admin)
Dashboard & Role Management
Login & Authentication
View execution history
Mark Outcome
Run
Reset test to active
Edit test case
Assign tester
View test result
Overview
Boards
Repos
Pipelines
Test Plans
Test plans
Progress report
Parameters
Configurations
Runs
Exploratory sessions
Artifacts
Project settings

Exp. 8 - Testing - Test Plans and Test cases.

Test plan - Cultural carnival Management System.

User stories

(1) As a user, I want to register and log in to access the System (ID :101)

(2) As a user, I want to view available events. (ID :102)

(3) As a user, I want to register for events (ID : 103)

(4) As a user, I want to book tickets for events.

(5) As an admin, I want to create and manage events (ID : 105)

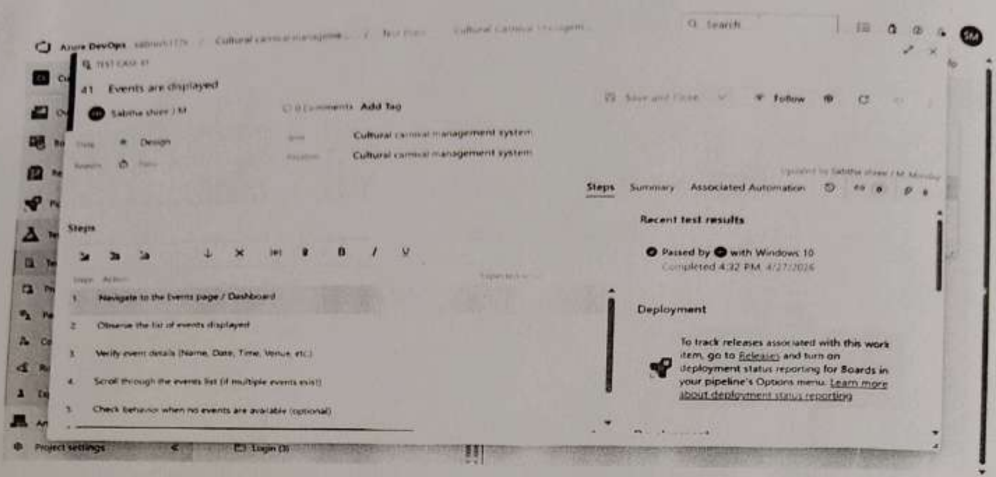
Test Suits & Test cases.

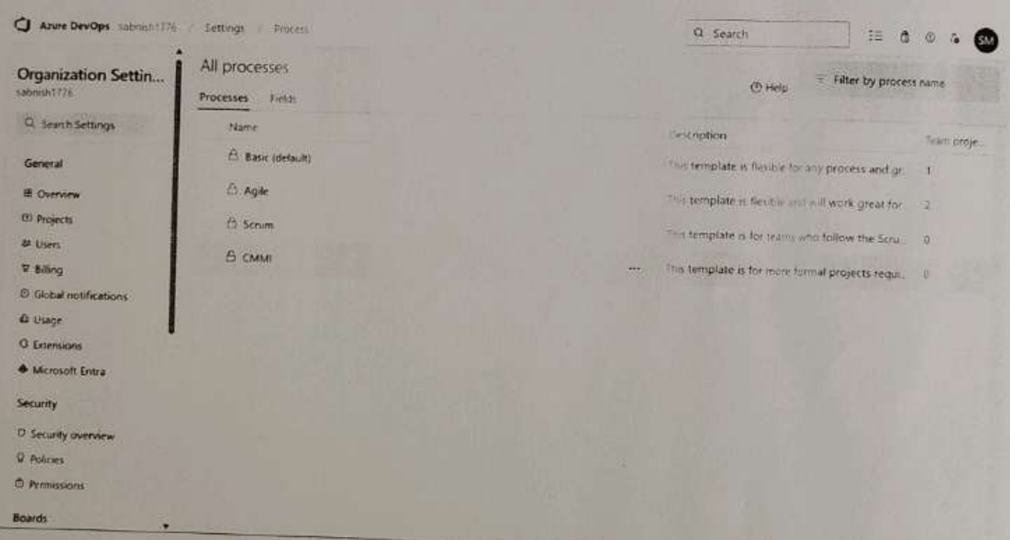
Test suite :- TS 01 - User Authentication (ID : 201)

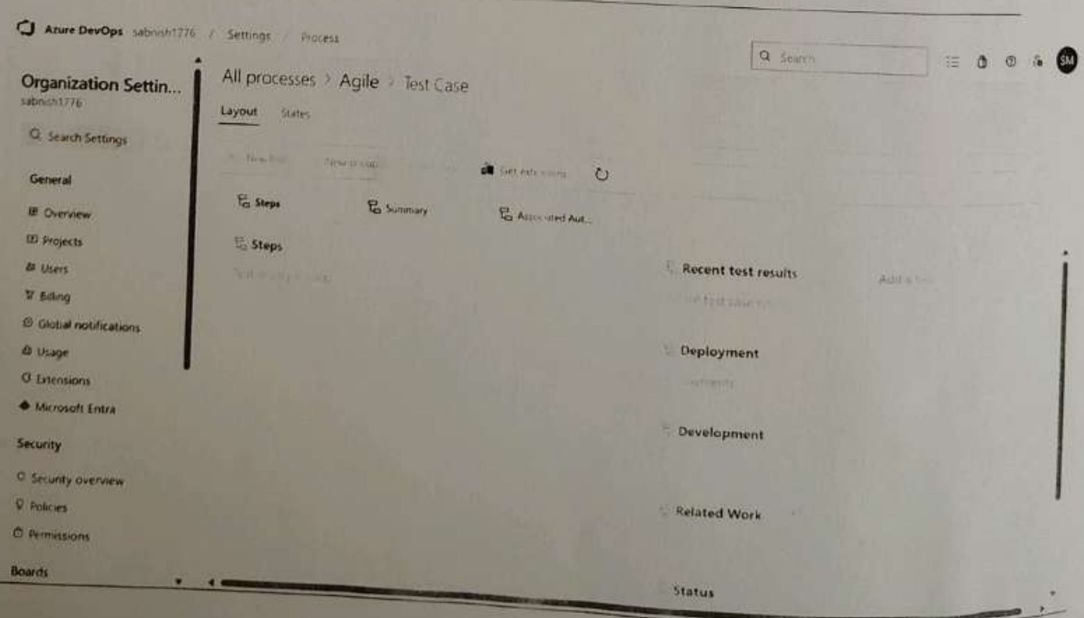
TC 01 - Successful user Registration.

Action : • Go to sign-up page
• Enter valid details.
• click "Register"

Expected Result.
• Account created successfully
• User redirected to dashboard.


Azure DevOps
Events are displayed
Add Tag
Cultural carnival management system
Follow
Steps
Summary
Associated Automation
Recent test results
Passed by with Windows 10
Completed 4:32 PM 4/27/2026
Deployment
1. Navigate to the Events page / Dashboard
2. Observe the list of events displayed
3. Verify event details (Name, Date, Time, Venue, etc.)
4. Scroll through the events list (if multiple events exist)
5. Check behavior when no events are available (optional)
To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting
Project settings


Azure DevOps sabrish1776 / Settings / Process
Search
Organization Settin...
sabrish1776
Search Settings
All processes
Processes
Fields
Help
Filter by process name
Name
Description
Team proje...
Basic (default)
This template is flexible for any process and gr... 1
Agile
This template is flexible and will work great for... 2
Scrum
This template is for teams who follow the Scru... 0
CMMI
This template is for more formal projects requi... 0
General
Overview
Projects
Users
Billing
Global notifications
Usage
Extensions
Microsoft Entra
Security
Security overview
Policies
Permissions
Boards


Azure DevOps sabrish1776 / Settings / Process
Organization Settin...
sabrish1776
Search Settings
All processes › Agile › Test Case
Layout
States
Steps
Summary
Associated Aut...
Steps
Recent test results
Deployment
Development
Related Work
Status
General
Overview
Projects
Users
Billing
Global notifications
Usage
Extensions
Microsoft Entra
Security
Security overview
Policies
Permissions
Boards

TC02 - login with valid credentials.

Action :

- Enter valid username & passwords
- click "login".

Expected Result:

- Account created successfully
- User redirected. to dashboard

Type : Happy path.

TC03 - Registration with existing Email.

Action :

Enter already registered email.
click "Register".

Expected Result:

- Error : "Email already exists".

Type : Error path.

TC04 : Login with Invalid password

Action :

- Enter wrong password

Expected Result :

- Error : "Invalid credentials".

Type : error path

Test Suite : TS02 – View Events (ID : 202)

TC05 – view event List

Action :

- login
- Navigate to Event Page.

Expected Result:

- List of all events displayed

Type : Happy path.

TC06 – Events loading failure.

actions :

- Turn off internet
- Open Events page.

Expected Result:

- Error : "Unable to load events.".

Type : Error path.

Test Suite : TS03 – Event Registration (ID : 203)

TC07 – successful Event Registration.

Action :

- select ~~ca~~ an event.
- click "Registeration"

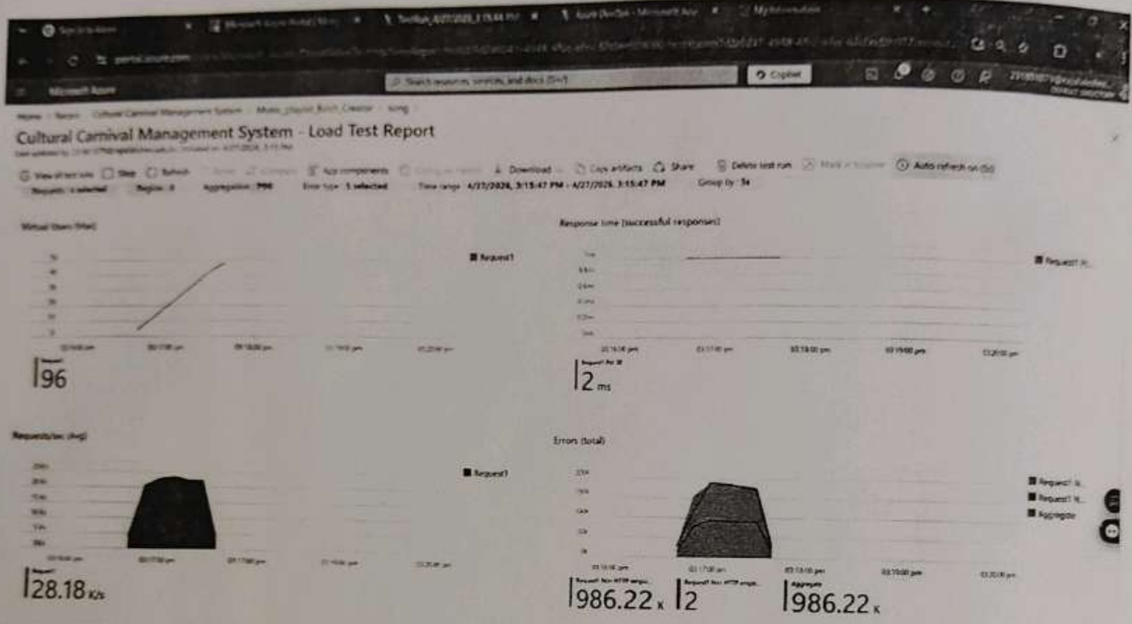
Expected Result:

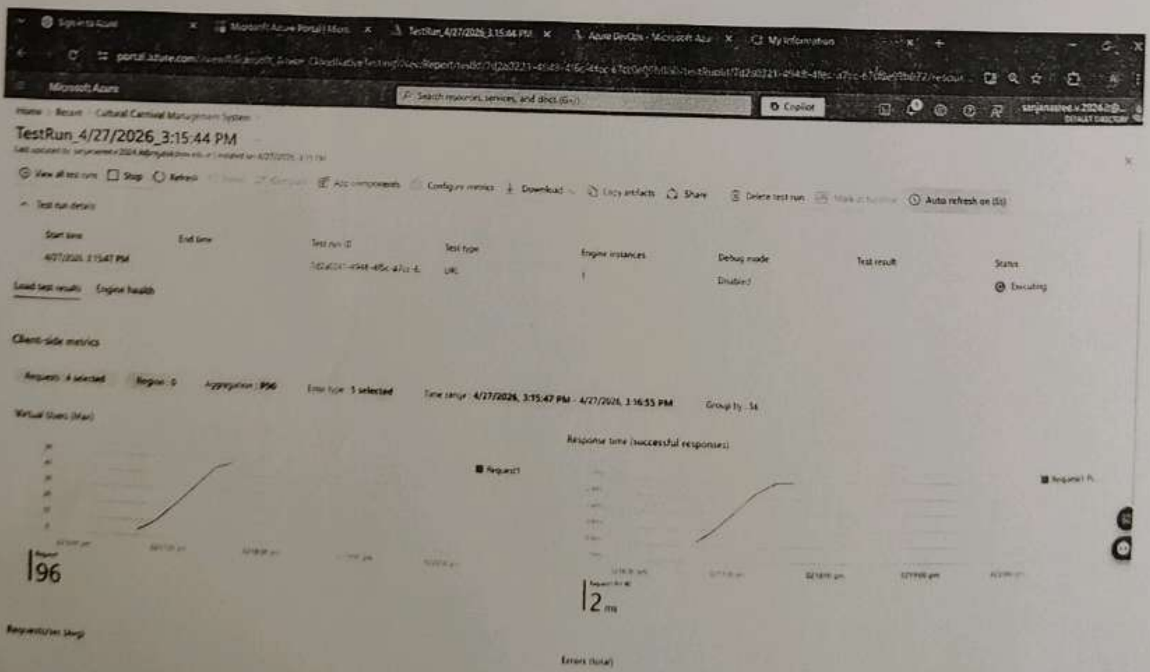
- Registration succesful message

Type : Happy Path.

EXP 9:

LOAD TESTING:





Exp 9: Load Testing and Pipelines.

Aim: To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an azure Devops pipeline for automating application builds tests and deployment.

Load Testing.

Azure load Testing:

Azure load Testing allows you to simulate high traffic and stress tests for your web applications and API's to understand how they perform under load. It helps identify performance bottlenecks, scalability issues and optimize resource usage before depoyment

→ Steps to create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure load Testing resources :

(1) Sign in to Azure Portal

Go to https://portal.azure.com and log in.

(2) Create the Resource

- Go to create a resource → search for "Azure Load Testing".
- Select Azure load Testing and click create

(3) Fill in the configuration Details.

- subcriptions : choose your Azure Subscription.
- Resource Group! create new or select an existing one
- name : provide a unique name (no special characters).
- location: choose the region for hosting the resource.

(4) (optional) Configure tags for categorization and billing
(5) click Review + create, then create.
(6) Once deployment is complete, click Go to resource.

→ steps to create and Run a load Test.

once your resource is ready

(1) Go to your Azure Load Testing resource and click Add Http requests > Create.

(2) Basics Tab

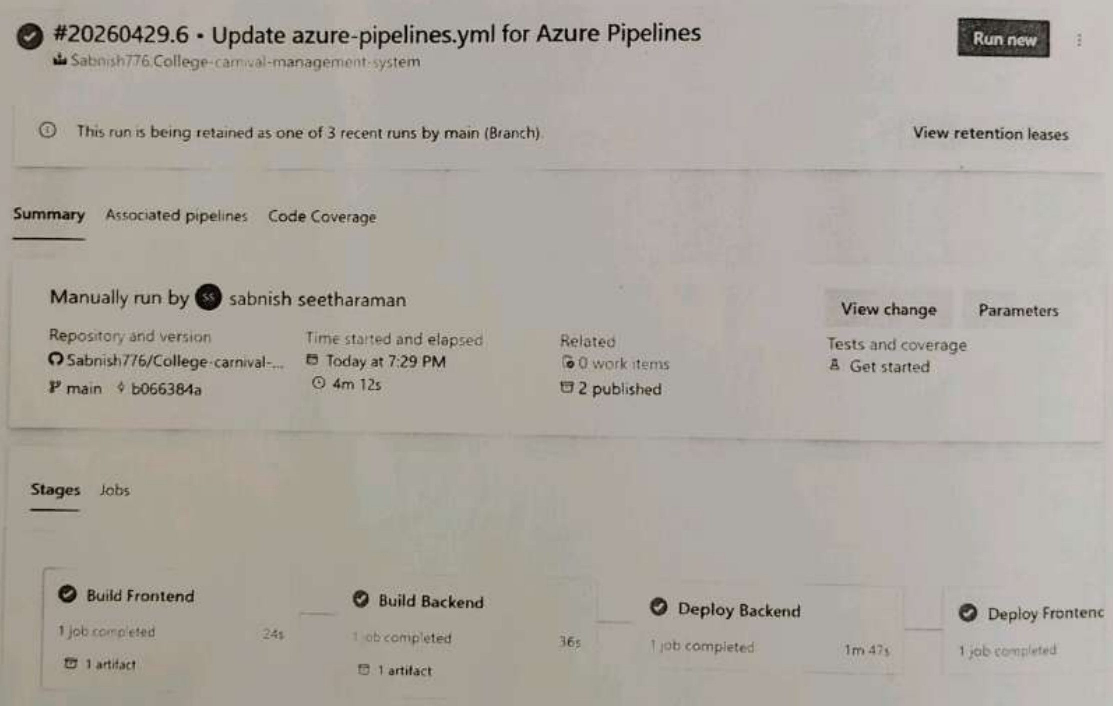
→ Test Name : provide a unique name.
→ Description : Add test purpose.
→ Run After creation : keep checked.

(3) Load settings.

→ Test URL : Target endpoint (eg: https://yourapi.com)

(4) ~~1.~~ click Review + create → create to start the test.

PIPELINES:



#20260429.6 • Update azure-pipelines.yml for Azure Pipelines
Sabnish776.College-carnival-management-system
Run new
This run is being retained as one of 3 recent runs by main (Branch).
View retention leases
Summary
Associated pipelines
Code Coverage
Manually run by sabnish seetharaman
View change
Parameters
Repository and version
Sabnish776/College-carnival-...
main
b066384a
Time started and elapsed
Today at 7:29 PM
4m 12s
Related
0 work items
2 published
Tests and coverage
Get started
Stages
Jobs
Build Frontend
1 job completed
24s
1 artifact
Build Backend
1 job completed
36s
1 artifact
Deploy Backend
1 job completed
1m 47s
Deploy Frontenc
1 job completed

Pipelines

Description

This experiment demonstrates how to connect Github-hosted ~~flask-based~~ spring boot based carnival management system project using azure Devops. This pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps :

(1) Connect Github to Azure Devops :

- In Azure Devops, create a new project
- Create a pipline and select Github as the source.
- Authorize access to your Github repository, ensuring that Azure Devops can pull the repository for your pipeline.

(2) Create azure-piplines.yml in your Root Repo.

- In your repository, create a new file called azure-pipelines.yml in the root directory
- Add the following basic pipeline configurations for react and springboot

(3) Pipeline Tasks Include

- Setting up the react and springboot environment
- Installing project dependencies from project ~~requ~~.

(4) Run and Monitor Pipeline

- Commit changes to the main branch of your repository to trigger the pipeline in Azure Devops.
- Monitor the logs in the Azure Devops portal to view logs, errors or success messages and ensure everything runs smoothly.

Result:

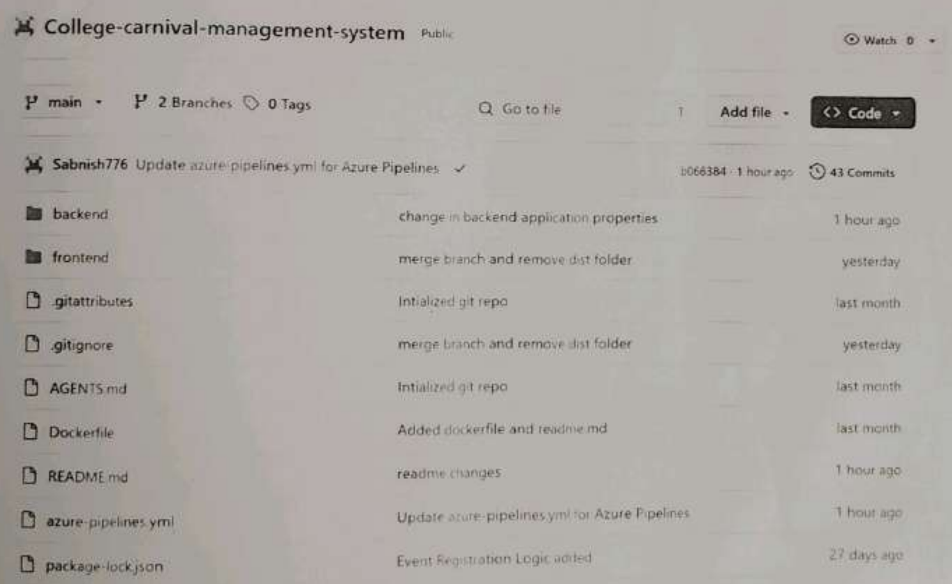
Successfully created the Azure load testing resource and executed a load test to access the performance of the specified endpoint and also demonstrated pipelines in azure devops.

Exp. 10 - Github structure & Naming conventions

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions helping contributors and users easily understand, navigate and extend the cultural carnival management System

EXP 10:



backend	change in backend application properties	1 hour ago
frontend	merge branch and remove dist folder	yesterday
.gitattributes	Intialized git repo	last month
.gitignore	merge branch and remove dist folder	yesterday
AGENTS.md	Intialized git repo	last month
Dockerfile	Added dockerfile and readme.md	last month
README.md	readme changes	1 hour ago
azure-pipelines.yml	Update azure-pipelines.yml for Azure Pipelines	1 hour ago
package-lock.json	Event Registration Logic added	27 days ago

Result:

The Github repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate.